

NETWORKS AND COMPLEXITY

Solution 15-10

*This is an example solution from the forthcoming book Networks and Complexity.
Find more exercises at <https://github.com/NC-Book/NCB>*

Ex 15.10: Heterogeneous Expansion [Lvl. 4]

Consider a network with N nodes and mean degree z . Existing links in this network are broken at a rate 1 per link. At rate r per node a node establishes a link to a randomly chosen partner.

- a) The dynamics of the degree distribution are captured by the differential equation

$$\dot{p}_k = -\underbrace{2rp_k}_1 - \underbrace{kp_k}_2 + \underbrace{2rp_{k-1}}_3 + \underbrace{(k+1)p_{k+1}}_4 \quad (1)$$

Explain the meaning of the terms 1-4.

Solution

The terms capture the following processes:

1. We lose nodes of degree k because they gain an additional link
 2. We lose nodes of degree k because they lose one of their links
 3. We gain nodes of degree k because nodes of degree $k-1$ gain a link
 4. We gain nodes of degree k because nodes of degree $k+1$ lose a link
- b) Define $G(x)$ as the generating function of the degree distribution. And derive a differential equation that captures the dynamics of G .

Solution

We define

$$G(x) = \sum p_k x^k \quad (2)$$

and write

$$\dot{G} = \sum \dot{p}_k x^k \quad (3)$$

$$= \sum (-2rp_k - kp_k + 2rp_{k-1} + (k+1)p_{k+1})x^k \quad (4)$$

$$= -2rG - xG' + 2rxG + G' \quad (5)$$

$$= 2r(x-1)G + (1-x)G' \quad (6)$$

- c) Consider the steady state of your equation and integrate to obtain $G(x)$. (Use the normalization conditions to fix the constant of integration.)

Solution

Setting the rate of change to zero yields

$$0 = 2r(x-1)G + (1-x)G' \quad (7)$$

Which we can also write as

$$G' = 2rG \quad (8)$$

and hence

$$G(x) = G(0)e^{2rx} \quad (9)$$

We don't know $G(0)$ straight away, but we know $G(1) = 1$ and hence

$$1 = G(0)e^{2r} \quad (10)$$

Solving for $G(0)$ yields

$$G(0) = e^{-2r} \quad (11)$$

which means

$$G(x) = e^{2r(x-1)} \quad (12)$$

- d) We have now found the generating function that the network approaches in the long run. What does it tell you about the network?

Solution

This is the generating function of an ER graph with $z = 2r$.